Lec1 Introduction to machine learning

Q1) What is the primary characteristic of supervised learning?

A) It uses labeled data to learn a mapping from input to output

Q2) which task involves predicting a contious numeric value?

B) Regression

Q3) which algorithm below is commonly used for clustering?

B) K-means

Q4) In reinforcement learning, what does the agent try to maximize?

B) Cumulative reward

Q5) In a confusion matrix, a false positive occurs when:

C) The model predicts positive but the actual class is negative

Q6) ⇒ C) Recall

Q7) TP = 90 FN = 10 FP = 5 TN = 895

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{90+895}{90+895+5+10} = \frac{985}{1000} = 98,5\%$$

C) 98,5%

Q8) ⇒ B) Mean squared Error (MSE